

This database design for the **Banking Management System** prioritizes data integrity, auditability, and performance as requested.

1. Database Overview

The schema is designed as a relational model to support core banking operations. It follows a modular structure: **Identity Management** (Users/Roles), **Core Banking** (Accounts/Transactions), **Lending** (Loan Applications), and **System Governance** (Audit/Fraud).

2. ER Diagram Description

- **Users** are the base entities with associated roles.
- **Accounts** belong to Users (1:N).
- **Transactions** link two accounts (Source/Destination).
- **Loan Applications** are tied to Users and managed by Employees.
- **Audit Logs** capture all sensitive changes across all tables.

3. Tables & Columns

Table	Primary Key	Foreign Keys	Key Columns
:--	:--	:--	:--
Users	user_id	-	username, password_hash, mfa_secret, kyc_status
Roles	role_id	-	role_name (Admin, Staff, Customer)
Accounts	account_id	user_id	balance, account_type, status, created_at
Transactions	txn_id	src_acc, dest_acc	amount, txn_type, status, fraud_score
Loans	loan_id	user_id	approved_by, amount, interest_rate, status, credit_score
Audit_Logs	log_id	user_id	action_type, table_affected, old_val, new_val

4. SQL Schema

```
CREATE TABLE Users (  
    user_id UUID PRIMARY KEY,  
    username VARCHAR(255) UNIQUE NOT NULL,  
    password_hash VARCHAR(255) NOT NULL,  
    mfa_secret VARCHAR(255),  
    role_id INT,  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);  
  
CREATE TABLE Accounts (  
    account_id UUID PRIMARY KEY,  
    user_id UUID REFERENCES Users(user_id),  
    balance DECIMAL(15, 2) DEFAULT 0.00,  
    status VARCHAR(20) DEFAULT 'ACTIVE',  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);  
  
CREATE TABLE Transactions (  
    txn_id UUID PRIMARY KEY,  
    src_account_id UUID REFERENCES Accounts(account_id),  
    dest_account_id UUID REFERENCES Accounts(account_id),  
    amount DECIMAL(15, 2) NOT NULL,  
    txn_status VARCHAR(20), -- PENDING, COMPLETED, FLAGGED  
    fraud_score INT,  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);  
  
CREATE TABLE Loans (  
    loan_id UUID PRIMARY KEY,  
    user_id UUID REFERENCES Users(user_id),  
    amount DECIMAL(15, 2) NOT NULL,  
    status VARCHAR(20), -- PENDING, APPROVED, DENIED  
    credit_score_at_time INT,  
    approved_by UUID REFERENCES Users(user_id),  
    updated_at TIMESTAMP  
);  
  
CREATE TABLE Audit_Logs (  
    log_id SERIAL PRIMARY KEY,  
    user_id UUID REFERENCES Users(user_id),  
    action_type VARCHAR(50),  
    table_affected VARCHAR(50),  
    timestamp TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

5. Indexes & Constraints

- **Indexes:**

- * ``idx_transactions_user``: Index on ``src_account_id`` and ``dest_account_id`` for fast statement generation.

- * ``idx_loan_status``: Index on ``status`` to allow employees to query pending loans quickly.

- * ``idx_audit_time``: Index on ``timestamp`` for compliance reporting.

- **Constraints:**

- * ``CHECK (balance >= 0)``: Ensures accounts cannot have a negative balance (for savings/checking).

- * ``CHECK (amount > 0)``: Ensures only positive transaction values.

- * ``FOREIGN KEY`` constraints are enforced strictly to maintain referential integrity across the banking lifecycle.

6. Normalization Notes

- **3NF (Third Normal Form):** The schema is normalized to 3NF to eliminate redundancy.
- **Denormalization Strategy:** While normally avoided, the ``balance`` in ``Accounts`` is a current-state column. Transaction history is kept separate to ensure the ledger remains immutable.
- **Auditability:** The ``Audit_Logs`` table uses a generic structure to ensure any DML (Data Manipulation Language) operation can be tracked without duplicating logic for every table.

7. Security Notes

- **Encryption:** The ``mfa_secret`` and ``password_hash`` are designed for integration with hardware security modules (HSMs).
- **Fraud Detection:** The ``Transactions`` table includes ``fraud_score``. This field is intended to be populated by the AI engine before a transaction moves from ``PENDING`` to ``COMPLETED``.